# A Comparison of Simulated and Physical Fluid Dynamics For Similitude Applications

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Abstract—Fluid dynamics simulations are useful when the full scale of a model or the environment it will be used in cannot be realistically experimented with. The purpose of this report is to compare fluid dynamics simulations with real-life measurements and prove that similitude can be used to apply measurements made on one system to describe the behaviour of similar systems. The results of this report can inform Harvey Mudd College E80 students whether simulating aquatic robots can be used to test model behaviour when in-field testing is not an option. To do this, drag forces and drag coefficients for a ⅓-scale model robot in a wind tunnel and a full-scale model robot in simulated water were obtained at specific fluid velocities and were subsequently compared. The measured and simulated drag forces versus Reynolds number followed predicted behaviour, where a parabolic relationship, confirmed through nonlinear regression, was found, and the measured drag forces were all smaller than the simulated drag forces. The measured and simulated drag coefficients were approximately the same for each corresponding Reynolds number, with an average percent difference of 7.29% including all data and 5.58% after removing an anomaly. Despite some discrepancies in the measured and simulated results due to differences in model behaviour, the concept of similitude was proven true: since the geometric and kinematic similarity conditions were true, the dynamic similarity condition could be applied, meaning the drag coefficients of the two tested systems were roughly the same, even with differences in individual variable values contributing to the drag coefficient.

## 1. Introduction

E80 is an experimental Engineering course at Harvey Mudd College in which students use a standardized robot to conduct experiments across various Engineering disciplines [1]. The purpose of this report is to compare the results of a COMSOL simulation of the full-scale E80 robot in water with real-life measurements of a ⅓-scale model of the E80 robot in air, and from this, prove that similitude can be used to apply measurements made on one system to describe the behaviour of similar systems. To do this, measurements of drag force, a resistive force from fluids on the object being modelled, and calculated from this, the drag coefficient, a proportionality constant in the equation for drag force, were recorded at fluid velocities chosen to yield equivalent Reynolds numbers, which represent the ratio of inertial and viscous fluid forces [2]. Drag coefficient and Reynolds number are dimensionless quantities, so if the two models have equal length ratios, their drag coefficients at equal Reynolds numbers should be the same.

This is important in E80 as students develop new robots for ocean deployment, and cannot test them in the ocean before deployment [1]. It is beneficial to compare fluid dynamics simulations with real-life measurements, since if data proving similitude is obtained, simulations are a reliable alternative for testing model behaviour if performing physical tests at the same scale and in the same environment is not an option.

The remainder of this report will present the methods used to conduct the physical and simulated experiments, including the physical experiment and simulation setup, data collection, and data processing. The results and discussion section will present data for the drag force and drag coefficients at various Reynolds numbers and discuss the conclusions drawn from this regarding similitude. The report will conclude by addressing the limits of experimentation and simulation and suggesting potential areas for future work.

## 2. Methods and Analysis

To measure the drag force on the ⅓-scale model E80 robot, a force measurement stand and a pitot tube with its end facing the direction of airflow were sealed in the rectangular test section of a wind tunnel using an acrylic panel and four spring clamps. The drag force in Kgf with zero velocity air flow was recorded from the force measurement stand. The temperature in °C, relative humidity as a percentage, and atmospheric pressure in Pa of the wind tunnel room were also recorded. The model robot was secured to the force measurement stand with a zip tie such that the protruding part of the robot faced the incoming wind. This was similarly sealed in the test section. While varying the maximum fan speed from 15-55% in increments of 5%, the differential pressure measured by the pitot tube and the drag force measured by the force measurement stand were recorded [3]. The drag force readings were tared by subtracting the zero velocity drag force reading from each model measurement. This was converted to Newtons for consistency with simulation values.

$$p_1 + 0.5\rho V_1^2 + \rho g z_1 = p_2 + 0.5\rho V_2^2 + \rho g z_2 \quad (1)$$

Equation (1) is Bernoulli's Equation, which relates pressure, velocity and elevation along a streamline, which is a line tangent to the velocity vector of the fluid at an instant, in nearly frictionless and incompressible flow [4]. In this equation, p is fluid pressure in Pa, ρ is fluid density in $kgm^{-3}$, V is fluid velocity in $ms^{-1}$, g is the gravitational constant in $ms^{-2}$, and z is height in m. The terms on the left side of (1) represent total values, and the terms on the right side of (1) represent values at a point on a streamline. In the wind tunnel, the change in ρgz is negligible, and the $0.5\rho V_2^2$ term is zero since fluid particles stop when they hit the pitot tube.

$$V_1 = \sqrt{\frac{2 \cdot \Delta p}{\rho}} \quad (2)$$

Equation (2) comes from rearranging (1) with the above substitutions. Δp is the differential pressure as reported by the pitot tube in Pa, ρ is the air density in the wind tunnel in $kgm^{-3}$, and $V_1$ is the air velocity in the wind tunnel in $ms^{-1}$. The relative humidity, atmospheric pressure, and air temperature were put into a calculator to obtain a value for air density [3]. The air density and differential pressure readings in the wind tunnel (converted from in $H_2O$ to Pa) were substituted into (2) to obtain values of the air velocity in each measurement.

$$Re = \frac{Ul}{\nu} \quad (3)$$

Simulations with the full-scale model required different fluid velocities to obtain equal Reynolds numbers as in the real-life trials. Equation (3) gives the formula for Reynolds number (Re), where U is the fluid velocity in $ms^{-1}$, l is the reference length in m, and ν is the kinematic viscosity of the fluid in $m^2s^{-1}$ [5]. The kinematic viscosity of air was $1.53\times10^{-5}m^2s^{-1}$ based on the wind tunnel temperature of 20°C, and the kinematic viscosity of water was $1\times10^{-6}m^2s^{-1}$ based on standard simulation values [6], [7]. Reference lengths that maintained the 1:3 ratio between the ⅓-scale model and full-scale model were used. By equating Reynolds numbers, corresponding simulated fluid velocities were found for each measured air velocity. Simultaneously, the Reynolds number for each fluid velocity in both the simulated and measured trials was recorded using (3).

A stationary study with laminar flow physics was set up in COMSOL Multiphysics® [7]. The solids and surfaces of the full-scale E80 robot were imported through a STEP file and modelled as acrylic plastic. A block, modelled as water, with dimensions large enough with respect to the robot model, was centred with the robot model to prevent boundary effects. The laminar flow inlet, with normal inflow velocity set to the calculated fluid velocities, and the outlet, with 0 pressure and backflow suppressed, were the faces of the block closest to the front (protruding section) and back of the robot model. The other four faces of the block were given symmetry. The wind tunnel was approximated as a coarse tetrahedral mesh, calibrated for fluid dynamics. The robot model was approximated as a coarse tetrahedral mesh, calibrated for general physics. To obtain drag force values, a surface integration of stress on the surface of the robot model for each fluid velocity was evaluated [3].

$$C_D = \frac{D}{0.5\rho U^2 A} \quad (4)$$

Equation (4) is the equation for the drag coefficient, where D is the drag force in N, ρ is fluid density in $kgm^{-3}$, U is fluid velocity in $ms^{-1}$, and A is the reference area in $m^2$. The reference area for (4) is the projected area - the area seen by an observer when looking at an object in the direction of fluid flow, which was $0.00164m^2$ for the ⅓-scale model and $0.01476m^2$ for the full-scale model [4]. The density of water used for simulated drag coefficients was $1000kgm^{-3}$ based on standard simulation values [7].

Reynolds number and drag coefficient are useful for determining if similitude conditions are fulfilled. Similitude is a method of applying measurements made on one system to another system. If the systems meet two similarity conditions, a third similarity condition must be true. This means if two nondimensional terms for one model equal the same nondimensional terms for another model, a third nondimensional term must be the same for the two models [4]. In this experiment, the geometric, kinematic, and dynamic similarity conditions were relevant. Two systems are geometrically similar when all model dimensions have the same linear scale ratio, all model angles and fluid flow directions are the same, and the models are orientated with respect to their surroundings in the same way. Two systems are kinematically similar if they have the same length and time scale ratios, which the Reynolds number encompasses per (3). Two systems have dynamic similitude if they have the same length, time, and force scale ratios, which the drag coefficient encompasses per (4) [4].

Plots of drag force versus Reynolds number and drag coefficient versus Reynolds number for the measured and simulated trials were produced and used to draw conclusions about similitude as described below.

**3. Results and Discussion**

Figure 1 shows a parabolic relationship between the drag force experienced by each robot model and the tested Reynolds number for the simulated and measured trials. The drag force has a direct squared relationship with fluid velocity squared according to (4), and the Reynolds number is proportional to fluid velocity according to (3). So, the drag force and Reynolds number should have a 2nd degree polynomial relationship, which matches the simulated and measured results. This was confirmed by performing a nonlinear least squares regression of Reynolds number versus drag force, where the 2nd order polynomial fit was the lowest order fit that showed fairly randomly scattered residuals about zero for both the measured and simulated data, as shown in Figure 1. The coefficients of the fit (in the order: quadratic, linear, and constant term) with 95% confidence bounds were $2.28\times10^{-11}$ ($9.28\times10^{-12}$, $3.64\times10^{-11}$), $8.02\times10^{-7}$ ($-2.33\times10^{-6}$, $3.94\times10^{-6}$), and -0.04 (-0.21, 0.12) for the measured data and $8.95\times10^{-11}$ ($8.85\times10^{-11}$, $9.05\times10^{-11}$), $1.76\times10^{-6}$ ($1.52\times10^{-6}$, $1.99\times10^{-6}$), and -0.05 (-0.06, -0.04) for the simulated data. The constant term confidence bounds for the measured data indicate

the polynomial fit could have a zero vertical offset, which aligns with expectations of zero drag force at zero fluid velocity. The constant term confidence bounds for the simulated data do not encompass zero, and instead indicate a negative drag force at zero fluid velocity. However, the predicted vertical offset is very small and can be attributed to errors in simulated drag force, as further explored in the discussion of drag coefficient results.

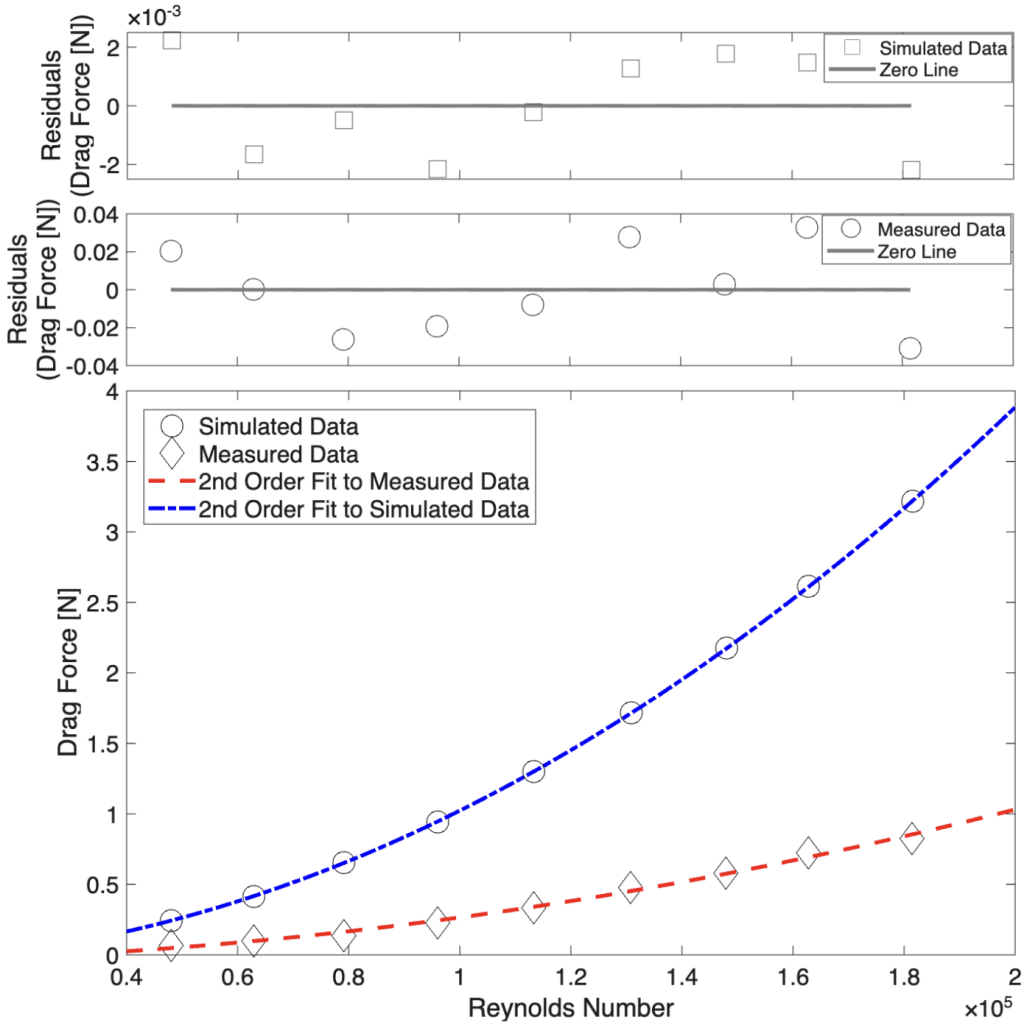



Figure 1. Plot of Residuals for Simulated Drag Force Data (top), Plot of Residuals for Measured Drag Force Data (middle), Drag Force Values at Reynolds Numbers Corresponding with Tested Fluid Velocities for the Full-scale E80 Robot Simulated in Water and the ⅓-scale E80 Robot Measured in a Wind Tunnel with 2nd Order Polynomial Fits (bottom). X-axis ticks indicating tested Reynolds Numbers are consistent for each plot.

Figure 1 also shows that all measured drag force values were smaller than the simulated drag force values. This matches predicted behaviour since the simulation used a larger scale of the E80 robot than in the real-life setup, meaning the reference area of the simulated model is greater. The simulation also used water as the flowing fluid, which is denser than air. This makes the simulated model experience a larger drag force than the physical model, since according to (4), drag force is proportional to fluid density and reference area.

Since the drag force values obtained were realistic, it was reasonable to derive drag coefficients from the data for comparison. Figure 2 shows that the values of the drag coefficient for the measured and simulated trials at each Reynolds number were about the same. They were all within the same magnitude, and only one trial yielded a percent difference between measured and simulated values greater than 10%. The measured and simulated drag coefficients at the Reynolds number of around $0.8\times10^5$ had a significant percent difference of 21.84%. This trial was most likely an anomaly, as the ⅓-scale model shifted in the wind tunnel, affecting the drag force it experienced and thus the drag coefficient reported. The average percent difference between measured and simulated drag coefficients was 7.29% with the anomaly and 5.58% without the anomaly, which is low, indicating the values are about the same. However, the percent difference in drag coefficients for the lowest four Reynolds numbers tested was over two times greater than that of the highest four Reynolds numbers tested. This was likely due to the range of Reynolds numbers used. While the exact ranges of Reynolds values that determine the type of flow of fluid around an object depend on flow geometry around the object and calculating

them would be out of the scope of this report, in general, low Reynolds numbers correspond with viscous creeping motion with negligible inertia effects, moderate Reynolds numbers correspond with laminar flow, and high Reynolds numbers correspond with turbulent flow [4]. The lower-end Reynolds numbers could have fallen in the range of viscous creeping motion, producing large discrepancies between the simulated and measured data since the COMSOL simulation was conducted assuming only laminar flow.

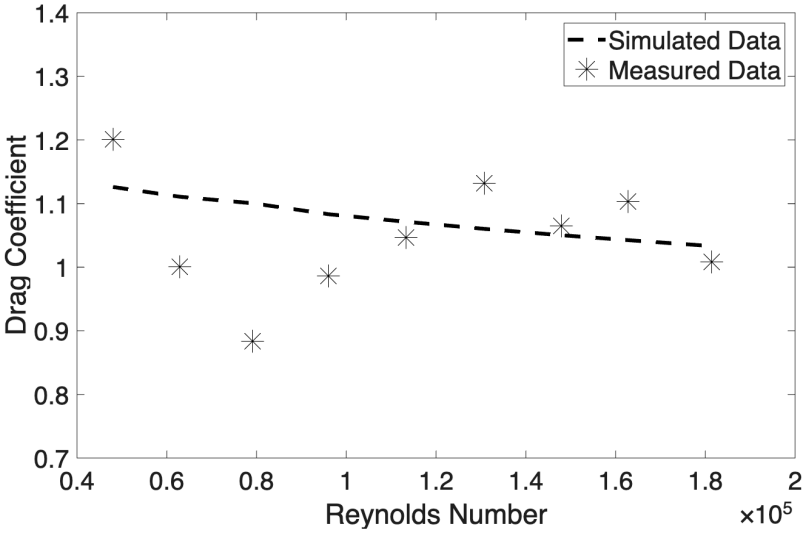



Figure 2. Drag Coefficient Values at Reynolds Numbers Corresponding with Tested Fluid Velocities for the Full-scale E80 Robot Simulated in Water and the ⅓-scale E80 Robot Measured in a Wind Tunnel.

Additional differences between the measured and simulated coefficient of drag results can be explained by discrepancies between real-life and simulated behaviour. As shown in Figure 3, COMSOL approximates modelled surfaces as meshes, whereas the ⅓-scale model had horizontal ridges and protruding strings. The coarse tetrahedral mesh used to approximate the E80 robot in COMSOL was very different from the smoothness of the ⅓-scale model. This changes the streamline of fluid flow around the models, which changes the drag force experienced by them. Meshes cut off convex geometry, shrinking the reference area, so simulated drag forces would be lower than in real-life, which could have produced the negative offset in the simulated polynomial fit [7]. Furthermore, COMSOL only calculates fluid flow at discrete points in space, as opposed to the continuous fluid flow in real life. It neglects anything smaller than its set feature size, producing errors in simulated drag force. In addition, zero velocity drag force readings, instead of drag force readings obtained from the equipment in the wind tunnel at each tested wind speed, were subtracted from each trial. The pitot tube and force measurement stand were in each trial and would have affected the drag readings to varying degrees at each wind speed, producing further discrepancies in the simulated and measured drag force. These drag force errors propagated, affecting the drag coefficient according to (4). Variations in drag coefficient values are more extreme at lower Reynolds numbers because the drag force values were tared using the zero velocity drag force readings. The difference between drag force readings and the zero velocity drag force readings decreases at lower wind speeds. Errors in either reading thus produce a larger percentage error at lower wind speeds, contributing to how the measured drag coefficients are closer to the simulated values at higher Reynolds number values in Figure 2.

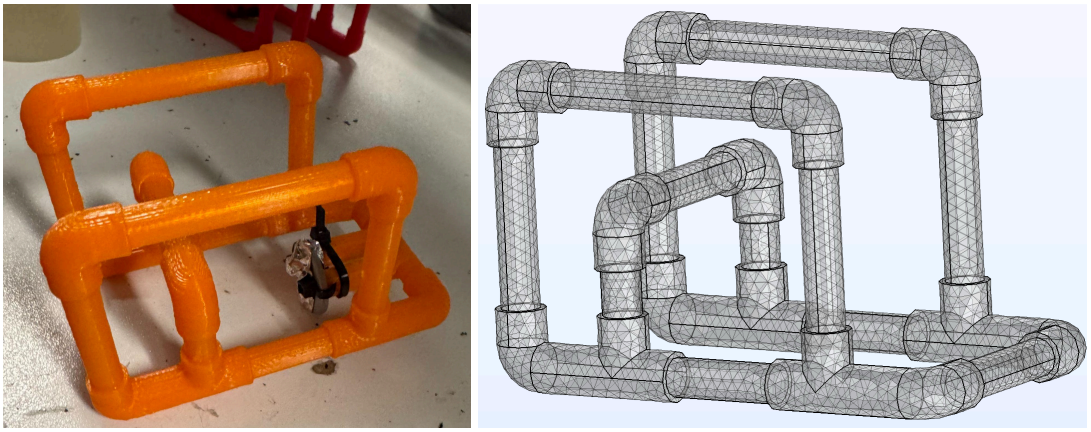


Figure 3. The ⅓–scale E80 Robot Model Used in the Physical Wind Tunnel (left) and the Full-scale E80 Robot Model Approximated as a Coarse Tetrahedral Mesh in COMSOL (right) [7].

Without the anomaly, all measured and simulated trials have reasonable percentage differences both as an average and individually (10% and under). Thus, the results align with predicted behaviour based on similitude. The ⅓-scale model in the wind tunnel and the simulated full-scale model fulfilled the geometric and kinematic similarity conditions. The two systems had geometric similitude because the robot models were scale versions of each other, and they were positioned in the physical and simulated wind tunnels such that fluid flow around them would be the same. The two systems were kinematically similar because the Reynolds numbers at each tested wind speed were equivalent for the two models. Since the geometric and kinematic similarity conditions were satisfied, the drag coefficients for the two models should have been roughly the same, which matches the obtained results. Therefore, the concept of similitude holds.

## 4. Conclusion

This report showed that by measuring a ⅓-scale model of the E80 robot in a wind tunnel and simulating a full-scale E80 robot model in water at various fluid velocities, the collected drag force and drag coefficient readings with respect to Reynolds number matched the predicted behaviour. Measured drag force values were all smaller than simulated values, and a parabolic relationship between drag force and Reynolds number was confirmed through nonlinear regression for both the measured and simulated data. The average percent difference between measured and simulated drag coefficients was 7.29% and 5.58% with and without the anomaly, and after disregarding the anomaly, all percent differences were under 10%. From these results, the concept of similitude can be applied to describe measurements on different systems with similar parameters. More broadly, the experimental and simulated data collection, data processing, and analysis procedures outlined in this report can be used to confirm similitude in other systems and find drag-related values in other geometries for predicting model behaviour when in-field testing is not feasible, such as for student-designed E80 robots.

Sources of error contributing to differences in simulated and measured drag coefficients include assuming laminar flow in simulation while using Reynolds numbers that could indicate differently, different streamlines of fluid and drag force experienced by each model due to distinct surface finishes, discretized approximations of simulated fluid behaviour that may not accurately represent continuous flow, and inaccurate measured drag force due to insufficient accounting for of pitot tube and force measurement stand.

Suggested future work to address these errors include calculating Reynolds number ranges for each type of flow and ensuring tested values fall within the laminar flow range, using a finer mesh approximation of the full-scale model in COMSOL and sanding the outside of the ⅓-scale model used in the wind tunnel to make their surface finishes more similar, shrinking the simulated feature size in COMSOL to approximate continuous flow better, and obtaining drag force readings for the pitot tube and force measurement stand at each measured wind velocity to tare the drag force readings more accurately.

## 5. References


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